Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science and Engineering (SCOPE)

Fall Semester 2024-25

Digital Assignment – 3

Course Code	Course Title	Class Type	Class ID	Slot	Registered Students
BCSE303P	Operating Systems Lab	LO	VL2024250102330	L39+L40	70

Maximum Marks: 10 **Due Date: 10th September 2024**

General Instruction(s):

- The program should be used by students, and they should take separate screenshots of the program and its execution.
- Student Name and Reg No. should be present in every program
- There shouldn't be any plagiarism or unethical behaviour in the code. Consider that if it is there, the mark will be reduced.
- Only the one document in PDF format should be uploaded by students. No marks will be given after the due date.

Q.No **Questions**

1. Consider a system with 5 processes (P0, P1, P2, P3, P4) and 4 resource types (A, B, C, D). The Allocation, Max, and Available matrices are given as follows:

	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	1	0	3	0	1	0	3	2	3	1	2
P1	1	0	0	1	2	2	2	2				
P2	1	3	2	0	3	3	2	1				
P3	1	2	1	2	1	2	2	2				
P4	0	0	1	1	2	1	1	2				

Using the Banker's Algorithm, answer the following questions:
1. Safety Check: Determine if the system is currently in a safe state. Show the sequence of processes and work vectors used to verify this.

2. Resource Request: Suppose process (P1) makes a request for resources: [1, 0, 1, 0]. Check if this request can be granted by:

- Verifying if the request is less than or equal to the process's remaining need.
- Verifying if the request is less than or equal to the available resources.
- Simulating the allocation and then performing a safety check to ensure the system remains in a safe state

(a) Write a C program to check whether the system is in safe state or not using Banker's algorithm? Also generate the safe sequence state.

(b) Compare the theoretical values with simulated results.

2. Write a C program and execute the implementation of classical problem of process synchronization using semaphores / monitors.
    - Bounded-Buffer Problem
    - Reader-Writer Problem
    - Dining philosopher's problem

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